SULIT

UTM
UNIVERSITI TEKNOLOGI MALAYSIA
Faculty of Computing

# UNIVERSITI TEKNOLOGI MALAYSIA
## FINAL EXAMINATION SEMESTER II , 2017 / 2018

| | | |
|---|---|---|
| **SUBJECT CODE** | : | **SCSJ 2013** |
| **SUBJECT NAME** | : | **DATA STRUCTURE AND ALGORITHM** |
| **SECTION** | : | **ALL** |
| **TIME** | : | **9.00 AM – 12.00 NOON** |
| **DATE/DAY** | : | **11 / 06 / 2018 ( MONDAY )** |
| **VENUES** | : | **BK 2 ( N 28 )** |

**INSTRUCTIONS :**

This examination book consists of **TWO ( 2 )** parts . **Answer ALL QUESTIONS.**
The layout of the questions are as follows:

| | | |
|---|---|---|
| **Part A :** | **TRUE / FALSE QUESTIONS** | **( 10 QUESTIONS :15 MARKS )** |
| **Part B :** | **STRUCTURED QUESTIONS** | **( 5 QUESTIONS : 85 MARKS )** |

**ANSWER ALL QUESTIONS IN THE ANSWER BOOKLET.**

**(Please Write Your Lecture Name And Section In Your Answer Booklet)**

| | |
|---|---|
| **Name** | |
| **I/C No.** | |
| **Year / Course** | |
| **Section** | |
| **Lecturer Name** | |

This questions paper consists of **NINE ( 9 )** printed pages excluding this page.

**PART A : TRUE/FALSE** **[15 marks]**

There are **10 Questions** in this section. Determine whether the statements given are **TRUE** or **FALSE**. Give justification for your answer.
( 1 mark for TRUE/FALSE answer and 0.5 mark for the justification.)

1. The complexity of Binary Search algorithm is $O(n \log_2 n)$. **TRUE/ FALSE**
   Justify your answer.

2. The worst case occurs in Linear Search algorithm when the searched item is the last element in the array or is not there at all. **TRUE/ FALSE**
   Justify your answer.

**Figure 1** shows 3 nodes and two pointers `ptr1` and `ptr2`. Answer Question 3 and 4 based on Figure 1.

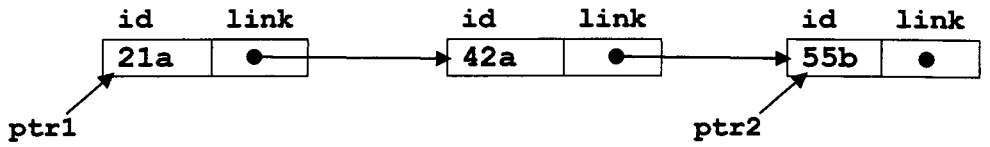



**Figure 1. Singly linked list**

3. The correct statement to print the `id` value for the second node in the linked list is: `cout << ptr1->id->link;` **TRUE/ FALSE**

   Justify your answer.

4. The Boolean expression: (`ptr2==NULL`) will return TRUE, because `ptr2` is currently points to the tail node (the last node in the list). **TRUE/ FALSE**
   Justify your answer.

5. 4 characters were inserted into a stack in the following sequences: **A B L E**. When pop operation is implemented to the stack until the stack become empty, the characters taken out from the stack are in the following sequences: **E L B A**. **TRUE/ FALSE**
   Justify your answer.

6. For array implementation of a stack, the following statements must be used in order to pop an item into the stack. **TRUE/ FALSE**

```
top = top - 1;
item = stack[top];
```

Justify your answer.

7. Both queue implementations of linear linked list and circular linked list require two pointers to access the front and back of the queue. **TRUE/ FALSE**
Justify your answer.

8. The source codes below is part of the codes to insert an item into a circular array queue. **TRUE/ FALSE**

```
front = (front + 1 % MAX_QUEUE)
count --;
```

Justify your answer.

9. In order to get the contents of a Binary search tree in ascending order; one has to traverse the tree in inorder traversal. **TRUE/ FALSE**
Justify your answer.

10. Every binary tree must have at least one node. **TRUE/ FALSE**

Justify your answer.

**PART B** **[85 marks]**

**THIS PART CONSISTS OF FIVE (5) QUESTIONS. ANSWER ALL QUESTIONS.**

**Question 1** **[15 marks]**

Based on the sorted array named **scores** shown in Figure 1a, answer the following questions:

| index | [0] | [1] | [2] | [3] | [4] | [5] | [6] | [7] | [8] | [9] | [10] |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2 | 5 | 11 | 17 | 29 | 37 | 43 | 47 | 69 | 74 | 88 |

**Figure 1a. scores** array

a) Perform basic sequential search for searching key **17** on the **scores** array in **Figure 1a**. Show the tracing of your search using variables **index**, **p**, and **found** as shown in the table format below: [3 marks]

| p | scores[p] | found |
|---|---|---|
| | | |
| | | |
| | | |

b) Perform binary search for searching key **17** on the **scores** array in Figure 1a. Show the tracing of your search using variables **left**, **right**, **middle**, **scores [middle]** and **found** as shown in the table format below: [4 marks]

| left | right | middle | scores[middle] | found |
|---|---|---|---|---|
| | | | | |
| | | | | |
| | | | | |

c) Based on your answer in Question a) and b), compare and discuss the performance of the two algorithms to search key 17 from **scores** array. [2 marks]

d) Fill in the following table with the number of comparisons and the complexity time required in searching the lowest score (**2**), average score (**37**) and highest score (**88**) on **scores** array in **Figure 1a** using Binary search and basic Sequential search technique. [6 marks]

| Searching Comparisons | | | | |
|---|---|---|---|---|
| **Search Key** | **Basic Sequential Search** | | **Binary Search** | |
| | **Number of Compares** | **Complexity** | **Number of Compares** | **Complexity** |
| 2 | | | | |
| 33 | | | | |
| 88 | | | | |

**Question 2** **[15 marks]**

a) Give two advantages of implementing linear list using linked list and one advantage of linear list using array. [4 marks]

`Program-2.1` is the implementation of `Position` class in C++. Answer questions 2(b) – 2(d).

```
1  // Program-2.1
2
3  class Position {
4      private:
5          float lat;  // lattitude
6          float lon;  // longitude
7          Position *next;
8          //.....
9      public:
10         Position(float pLat, float pLon) {
11             lat = pLat; lon = pLon; next = NULL;
12         }
13
14         void printPosition(void) {
15             cout << "(" << lat << ", " << lon << ")" << endl;
16         }
17 };
```

b) Given the following codes and based on the `Position` class declaration in Program 2.1, draw diagrams that represent the node pointed by **P1** and **P2**.

```
Position *P1 = new Position(1.54,103.62);
Position *P2 = new Position(1.57,103.64);
```

[3 marks]

c) Draw linked list diagrams for the question i) and ii) that show the execution of the following codes.

i) `Position *head = P1;` [2 marks]

ii) `P2->next = P1;`
`head = P2;` [3 marks]

d) Assume that the linked list structure of `Position` instances have been successfully constructed in Question 2 (c), write source codes to delete the node pointed by **head** and **P2** and print message :

```
Node with longitude 1.57 and latitude 103.64 has been deleted
```

[3 marks]

**Question 3** **[20 marks]**

a) Convert the infix expression `( A + 5 * B / C ) - 2 * D` into prefix and postfix expressions by following the precedence, association and parentheses rules. Show the steps.

[5 marks]

Given Algorithm 3.1, answer Question 3-b

```
//Algorithm 3.1 - check parentheses

create(stack);
continue=true;
while ((not end of input expression) && continue){
      ch=getch();
      if ((ch=='(') || (ch==')')){
            if (ch=='(')
                  push(stack, '(');
            else if (isEmpty(stack))
                  continue=false;
            else
                  pop(stack);
      }//endif
}//endwhile
if (end of input && isEmpty(stack))
      cout<<"the parentheses is balance";
else
      cout<< "the parentheses is not balance";

//end of algorithm
```

b) Based on Algorithm 3.1 for checking the balance parentheses using stack, show that the expression `((P - Q) / ( ( R + S * T )` is unbalance. Show the checking process by completing the table below:

| Input Ch | Content of Stack |
|---|---|
| | |
| | |

[5 marks]

c) Show the steps of evaluating the postfix expression **5 4 * 2 / 3 2 + - 6 +**. Use the format given in the table below in order to show the evaluation process.

[5 marks]

| postfix | ch | opr | opnd1 | opnd2 | result | stack |
|---|---|---|---|---|---|---|
| | | | | | | |
| | | | | | | |

Given the declaration of **Stack** class in Program-3.1, answer Question d).

```
1  // Program-3.1
2  // Stack class declaration
3  class Stack
4  {
5      private:
6          int top;
7          char data[MAXSTACK];
8
9      public:
10         Stack();
11         bool isFull();
12         bool isEmpty();
13         void push(char);
14         char pop();
15         void printLIFO();
16         void printPop()
17 };
```

d) Complete the code for **pop()** function so that it will return the popped item (**char**) if the stack is not empty and return an asterisk '*' if the stack is empty.

[5 marks]

**Question 4** **[15 marks]**

a) Give the differences between the implementation of the queue implementation linear array and queue implementation circular array in the following operations:
   i) Queue declaration
   ii) **enQueue()** function.
   iii) **deQueue()** function.

[3 marks]

b) Figure 4a shows the current content of a circular queue.

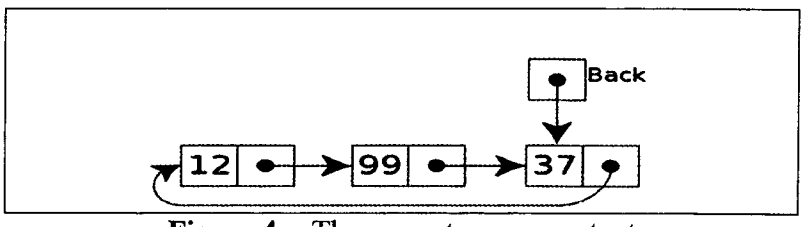



**Figure 4a:** The current queue content

Redraw the queue in Figure 4a that shows the current content of the queue after the sequence of instructions in Figure 4b is executed.

```
Queue -> deQ();
Queue -> enQ(30);
Queue -> deQ();
```

**Figure 4b :** Queue instructions

[3 marks]

Answer Question d), which deals with operations on queue implementation array. Figure 4c shows Queue 1 with maximum content of **5** items in the queue. Currently, there are three items already in the queue.

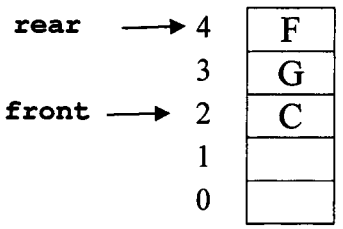



**Figure 4c:** Queue 1

d) Perform the following tasks to Queue 1 using circular array:

i. Redraw Queue1 after 1 deletion and 3 inserts have been performed: `enQueue(H)`, `deQueue()`, `enQueue(I)` and `enQueue(J)`. Label the correct location for `front` and `rear`. [3 marks]

ii. In this situation, give the condition that indicates the queue is full.

[2 marks]

e) After a series of **enQueue** and **deQueue** operations in question (d), discuss the efficiency of memory usage for the approaches if compared with queue using linear array. What problem does this approach solve?

[4 marks]

**Question 5** **[20 marks]**

Answer Questions a) based on the expression tree in **Figure 5a**.

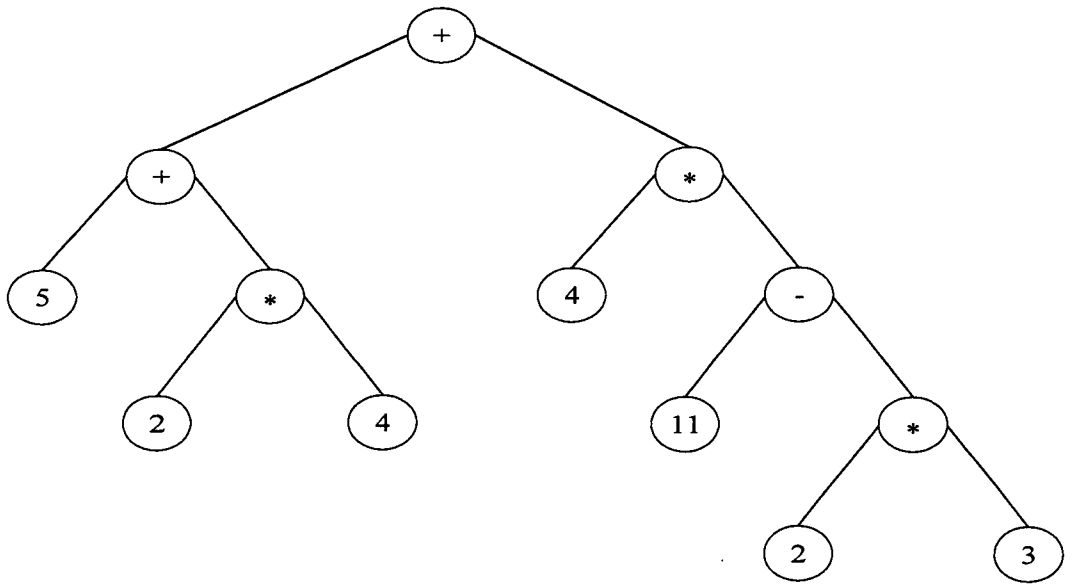



**Figure 5a:** Expression tree

a) From the tree, list the following:

i) Height of the tree

ii) Level of node 5

iii) Ancestor of node 11

iv) The tree characteristics (balance/full/complete)

v) Give the inorder, preorder and postorder of the tree traversal.

vi) What is the result of the expression?

[11 marks]

b) Draw a binary search tree based on the following sequence of input:

**F I N A L D S O K** (5 Marks)

Answer Questions c) based on the binary search tree in **Figure 5b**.

c) Given the binary search tree in Figure 5b, redraw the tree after the following nodes are deleted **in sequence**. Show the new tree after every deletion.

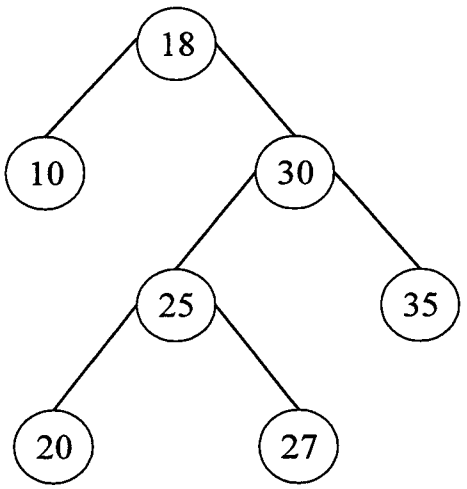



**Figure 5b:** Binary search tree

(a) Delete node 25 [2 marks]

(b) Delete node 18 [2 marks]